The Islamia University of Bahawalpur

Final Year Project Proposal

Program: ☐ BSSE ☆ BSIT ☐ BSAI

Project Title: Fake Profile Detection

Nature of Project: ☆Development ☐ Research

Area of Specialization:

☆ Artificial Intelligence (AI)	☐ Game Development	☐ Internet of Things (IoT)
☆ Machine Learning	☐ Mobile App Development	☐ Blockchain
☐ Data Science and Analytics	☆ Web Development	☐ Cybersecurity
☐ Image Processing	☐ Desktop Application	
☐ Natural Language Processing (NLP)	☐ Other (Specify)_____	

Introduction:

Fake Profile Detection is a web-based system that checks the public details of an Instagram profile and predicts whether the account is real or suspicious. It uses a trained machine learning model to give a risk score and explain why a profile may look suspicious.

Instagram is used by millions of people for communication, shopping, and promoting businesses. However, some people create fake accounts for fraud, spam, harassment, pretending to be someone else, and increasing follower counts unfairly. Many users find it difficult to know whether an unknown account is trustworthy.

This project aims to provide a simple tool to help users check Instagram profiles. Users enter visible profile details, such as the number of followers and following, number of posts, username pattern, biography, and profile picture status. The system analyzes these details and shows a result with a simple explanation.

This project helps improve online safety and trust while showing how machine learning can be used to solve real-world problems.

Problem Statement :

Fake Instagram profiles are created in large numbers, and ordinary users often find it difficult to identify them. These accounts may be used for fraud, spam, harassment, impersonation, and other harmful activities. Previous research has explored machine learning techniques to detect fake Instagram accounts (Chelas et al., 2024; Azami & Passi, 2024). However, detecting fake accounts remains challenging because the results depend on the available data, selected profile features, and performance of the machine learning model.

The following limitations are important to consider in existing methods:

1. Time-consuming manual checking: Users must examine profile details themselves, which takes time and may lead to incorrect judgments.
2. Difficulty identifying suspicious accounts: Some fake accounts may look similar to genuine accounts, making them difficult to identify by appearance alone.
3. Dependence on data quality: Machine learning models depend on the quality and type of data used for training. Limited or unsuitable data may affect their performance (Chelas et al., 2024).
4. Lack of understandable results for ordinary users: Research models may focus on classification accuracy, while ordinary users also need simple explanations to understand why a profile is considered suspicious.

To address these problems, this project proposes a web-based Fake Instagram Profile Detection System using machine learning. Users will enter publicly visible profile details, such as the number of followers and following, number of posts, username pattern, biography, and profile picture status. The system will analyze the available details using a trained machine learning model and provide a prediction with a risk score and an explanation of the result. The aim is to help users make more informed decisions about unfamiliar Instagram accounts. The system will support the identification of potentially suspicious profiles, but it will not guarantee that every prediction is correct.

References

1. Chelas, S., Routis, G., & Roussaki, I. (2024). Detection of Fake Instagram Accounts via Machine Learning Techniques. Computers, 13(11), Article 296. https://doi.org/10.3390/computers13110296
2. Azami, P., & Passi, K. (2024). Detecting Fake Accounts on Instagram Using Machine Learning and Hybrid Optimization Algorithms. Algorithms, 17(10), Article 425. https://doi.org/10.3390/a17100425

Project Goal / Objectives:

Goal: To build a web-based system that predicts whether an Instagram profile is real or suspicious and explains the result in a way any user can understand.

Objectives:

1. Collect and prepare a labeled dataset of Instagram profile features (followers, following, posts, username pattern, biography, profile picture, account privacy).
2. Train and compare at least three classification algorithms (Logistic Regression, Decision Tree and Random Forest) using Scikit-learn.
3. Reach at least 90% accuracy on a held-out test set, and report precision, recall and F1-score alongside it.
4. Calculate a risk score from 0 to 100 for every checked profile and label it Low, Medium or High risk.
5. Show the main reasons a profile was marked suspicious, for example an unusual followers-to-following ratio or digits in the username.
6. Provide a Flask web interface with a profile check form and a results page.
7. Store every scan in an SQLite database and let the user view the scan history.
8. Return a result within 3 seconds for a single profile check.

Application Architecture / Technology Stack / Tools / Hardware :

Programming Languages: Python, HTML, CSS, JavaScript.
Architecture: Client–Server Architecture using Flask, with separate frontend, backend, database, and machine learning components.
Frameworks and Libraries: Flask for web application development; Scikit-learn for implementing the machine learning model.
Database: SQLite for storing application-related records.
Tools: Visual Studio Code for coding, Python for development, and a web browser for testing the application.
Machine Learning: A classification model will be selected and trained using a suitable labelled dataset to identify potentially fake social media profiles.
Hardware: A laptop or desktop computer with an Intel Core i3 or equivalent processor, at least 4 GB RAM, and sufficient storage space.

Project Group Members:

Sr.#	Reg. #	Student Name	Phone #	Signature
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(2)				
(3)				

References (if any):

1. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825‑2830.
2. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5‑32.
3. Cresci, S., Di Pietro, R., Petrocchi, M., Spognardi, A., and Tesconi, M. (2017). The paradigm-shift of social spambots: Evidence, theories, and tools for the arms race. Proceedings of the 26th International Conference on World Wide Web Companion.
4. Flask Documentation, Pallets Projects: flask.palletsprojects.com

Project Supervisor:

Name: Dr. Rana Nazeer Ahmad Supervisor Acceptance Signature: _____

Head of Department (HoD): _____